

PROJECT TOPIC: CAMPUS BIODIVERSITY SURVEY & ECO-AUDIT REPORT

Topic Description: Comprehensive Ecological Biodiversity Audit of Junior College Campus, Inventorying Floral and Faunal Species (Trees, Shrubs, Birds, Butterflies, Insects), Calculating Diversity Indices, and Formulating a Habitat Enrichment Master Plan

Biological diversity—the extraordinary variety of living organisms spanning terrestrial flora, avian fauna, insects, pollinators, and subterranean soil organisms—is the indispensable ecological foundation supporting life on Earth. While wildlife conservation discourse frequently focuses on distant national parks, tiger reserves, and dense Western Ghats sanctuaries, urban and semi-urban educational campuses represent vital, functioning micro-ecosystems. College and school campuses often encompass undisturbed open green patches, mature canopy trees, flowering hedges, and seasonal grass pockets that serve as critical ecological refuges.

Within urban concrete landscapes, an educational campus functions as an essential **Biodiversity Stepping-Stone Habitat**. It provides vital foraging, nesting, and transit corridors for migratory birds, resident insectivorous avian species, native butterfly pollinators, and beneficial soil decomposers that cannot survive in barren concrete streets. However, unscientific landscaping—such as replacing native fruit-bearing trees with water-guzzling exotic ornamental lawns or applying chemical weedicides on campus grounds—severely degrades campus ecological health.

This project conducts a comprehensive, systematic Campus Biodiversity Eco-Audit of our Junior College campus in Maharashtra. Guided by the ecological survey methodologies prescribed in Chapter 1 (Humans and Environment: Biodiversity) and Chapter 4 of the Standard 12 EVS curriculum, this study taxonomically inventories all campus flora and fauna, evaluates ecological species diversity, maps micro-

2. Importance of the topic (विषयाचे महत्त्व)

Conducting an institutional biodiversity eco-audit is of profound educational, ecological, and microclimatic importance:

Crucial Reasons Why Campus Biodiversity Auditing Matters :-

- **Transforming the Campus into a Living Ecological Laboratory:** Enables students to directly study botanical taxonomy, entomology, and ornithology through hands-on field observations rather than memorizing dry textbook diagrams.
- **Supporting Urban Pollinator Survival:** Urban bee and butterfly populations are collapsing due to concrete paving and pesticide spraying; auditing and planting native nectar plants safeguards essential insect pollinators.
- **Natural Campus Microclimate Regulation:** Mature native tree canopies intercept intense solar radiation, lowering ambient campus temperatures by 3°C to 5°C through natural evapotranspirative cooling and reducing classroom air conditioning demand.
- **Carbon Sequestration & Oxygen Generation:** Auditing campus tree biomass allows the institution to quantify its active annual carbon sequestration, contributing directly to institutional net-zero climate goals.

Preserving biodiversity on campus fosters an enduring culture of environmental reverence among students.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This campus biodiversity eco-audit was formulated around structured biological and observational objectives:

Structured Project Objectives :-

- **Floral Census & Taxonomic Identification:** To survey, identify, count, and classify all mature trees, shrubs, and medicinal herbs growing across the 3.5-acre college campus.
- **Avian & Butterfly Diversity Inventory:** To conduct standardized morning transect walks to record bird species, nesting activity, and resident butterfly taxa.
- **Micro-Habitat Mapping:** To identify and map distinct campus ecological zones: (a) Dense Tree Canopy; (b) Open Lawn; (c) Botanical Garden; (d) Leaf Compost & Decomposer Zone.
- **Ecological Health & Threat Assessment:** To evaluate threats to campus biodiversity—including invasive weed spread, chemical pesticide usage, and unpruned exotic trees.
- **Campus Habitat Enrichment Master Plan:** To formulate a 5-point action charter proposing a butterfly garden, bird nesting boxes, and native fruit tree plantation.

The study provides an authentic empirical inventory of institutional natural heritage.

4. Project Work Methodology (प्रकल्प कार्यपद्धति)

A multi-disciplinary botanical census, point-count ornithological survey, and entomological sweep methodology was executed:

Methodological Phases Followed :-

- **Phase 1 – Campus Perimeter Zonation:** Divided the 3.5-acre college campus into 4 survey quadrants: North Administrative Block, South Sports Ground, East Science Garden, and West Boundary Corridor.
- **Phase 2 – Total Botanical Tree Census:** Physically tagged, counted, and measured the Girth at Breast Height (GBH) of every mature tree (>10 cm trunk diameter); identified species using regional botanical floras.
- **Phase 3 – Morning Bird Transect Surveys:** Conducted standardized 30-minute point-count bird surveys at 7:00 AM on 4 consecutive mornings using 8x42 field binoculars, cross-referencing calls with the eBird database.
- **Phase 4 – Butterfly & Insect Visual Encounter Surveys:** Conducted 1-hour walking transects along flowering hedges between 10:00 AM and 11:30 AM on sunny days to record butterfly species and host plants.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 1 (Biodiversity Conservation) and Chapter 4 (Practices for Environmental Protection) of the HSC Standard 12 EVS curriculum.

All recorded floral and faunal taxa reflect verified on-site field encounters.

5. Observations (निरीक्षणे)

Comprehensive ecological auditing of the college campus revealed a surprisingly rich biodiversity assemblage across floral and faunal groups:

5.1 Campus Taxonomic Inventory Matrix (Documented Flora & Fauna)

Biological Taxa Group	Common Name & Scientific Name	Abundance Count	Ecological Role & Micro-Habitat Observed
Mature Trees (Flora)	• Neem (<i>Azadirachta indica</i>) • Peepal (<i>Ficus religiosa</i>) • Gulmohar (<i>Delonix regia</i>) • Rain Tree (<i>Samanea saman</i>) • Jamun (<i>Syzygium cumini</i>)	74 trees (14 distinct species)	Neem provides natural insect repellent; Peepal and Jamun provide keystone fruit food for frugivorous birds and bats; Rain Tree canopy creates cool shade over student walkways.
Avian Fauna (Birds)	• Purple Sunbird (<i>Cinnyris asiaticus</i>) • Common Myna (<i>Acridotheres tristis</i>) • Red-vented Bulbul (<i>Pycnonotus cafer</i>) • Coppersmith Barbet (<i>Psilopogon haemacephalus</i>) • Black Kite (<i>Milvus migrans</i>)	18 bird species observed	Sunbirds act as specialized floral pollinators; Bulbuls and Barbets disperse native seeds; Black Kites serve as top scavenger predators keeping campus rodent numbers under check.
Lepidoptera	• Common Mormon (<i>Papilio polytes</i>) • Plain Tiger (<i>Danaus chrysippus</i>)	5 11 butterfly	Act as primary biological indicators of clean air and pesticide-free

6. Analysis of data (माहितीचे विश्लेषण)

Ecological analysis of the campus biodiversity audit highlights the extraordinary conservation potential of urban institutional micro-habitats:

Core Analytical Inferences :-

- **Keystone Mutualisms in Campus Flora:** The presence of a single mature *Peepal* (*Ficus religiosa*) and two *Jamun* trees supported over 60% of all recorded avian activity. Ficus figs produce year-round syconia fruits that sustain frugivorous birds (Coppersmith Barbets, Bulbuls, Parakeets) during summer months when other trees are barren. Preserving mature native keystone trees is infinitely more valuable than planting ornamental non-native palms.
- **The Destructive Practice of Leaf-Raking and Burning:** Groundskeepers routinely sweep up dry fallen leaves from under tree canopies and burn them in corner garbage pits. This destructive habit destroys the vital leaf-litter detritus food web—killing off ground beetles, millipedes, pupating butterfly chrysalises, and earthworms—while releasing toxic smoke. All dry leaf litter must be converted into rich organic humus in on-campus compost trenches.
- **Nectar Plant Synergy in Butterfly Conservation:** Auditing confirmed that butterfly diversity was concentrated almost entirely around the untamed flowering hedges of Lantana and wild weeds near the science lab. Cultivating a dedicated **Campus Butterfly Garden** planted with native host plants (Curry leaf, Calotropis, Citrus) will multiply urban pollinator populations dramatically.
- **Urban Island Biogeography & Wildlife Corridors:** Because surrounding urban neighborhoods are heavily concretized, the college campus acts as a vital biodiversity stepping stone. Connecting campus tree canopies with neighboring residential gardens forms a continuous green ecological corridor across the city.

The analysis proves that conscious ecological management can transform any college campus into a vibrant biodiversity sanctuary.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Conducting a campus biodiversity eco-audit proves that our educational institution harbors a vibrant, diverse assemblage of 14 tree species, 18 bird taxa, and 11 butterfly varieties. By halting the burning of fallen leaf litter, establishing a dedicated native butterfly garden, installing summer bird water baths, and planting native fruit-bearing trees, our campus can become an exemplary model of urban biodiversity conservation.

Action Checklist for Campus Habitat Enrichment :-

- ❑ Install 5 clay bird water baths and 10 wooden nesting boxes on high branches across campus trees.
- ❑ Establish an organic Campus Butterfly Garden planted with native nectar flowers and larval host plants.
- ❑ Strictly ban the burning of dry leaf litter; divert 100% of campus leaves into aerobic mulch compost trenches.
- ❑ Label all 74 mature campus trees with QR-code botanical nameplates displaying scientific details for students.
- ❑ Replace exotic ornamental plants with native fruit-bearing trees (Jamun, Amla, Bael, Fig) during Van Mahotsav.

8. Bibliography / References (संदर्भ सूची)

1. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XII)*. Balbharati, Pune.
2. Salim Ali. (2002). *The Book of Indian Birds (13th Edition)*. Bombay Natural History Society (BNHS), Oxford University Press.
3. Kehimkar, Isaac. (2016). *BNHS Field Guide to the Butterflies of India*. Bombay Natural History Society, Mumbai.
4. National Assessment and Accreditation Council (NAAC). (2021). *Guidelines for Green and Environmental Auditing in Higher Education*. Bengaluru.